

Hongcheng Jiang

Applied AI Engineer | Agent Systems

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Kansas City, MO | Open to relocation / remote | U.S. Permanent Resident (no sponsorship required)

SUMMARY

Applied AI engineer building tool-using LLM agents for SQL analysis and medical measurement. Implements bounded execution, repair loops, durable asynchronous workflows, and failure recovery, with reproducible evaluation and human review.

TECHNICAL SKILLS

Agent Systems: Tool Calling, MCP, Structured Outputs, Execution Control, Bounded Repair, Agent Evaluation

Engineering: Python, SQL, FastAPI, SQLite, Asynchronous Workers, Docker, Linux, Git

ML: PyTorch, Hugging Face, vLLM, Ollama, Computer Vision, Diffusion Models, Vision Transformers

PROFESSIONAL EXPERIENCE

University of Missouri-Kansas City

Jun 2026 - Present

Research Associate (Postdoctoral Researcher)

Kansas City, MO

- RadMeasure — Medical Measurement Agent: Constrained LLM-proposed actions to registered protocols and tools; implemented geometry verification and a **KEEP/REPAIR/STOP controller** with repair budgets, independent-proposal checks, human review, and execution traces.
- RadMeasure repair evaluation: Across 528 runs on 176 archived cases, intervened in 106 runs (20.1%) with **0.69° mean error reduction among intervened runs**; overall MAE fell from 2.68° to 2.54°.
- GeoVIRA (submitted to WACV 2027): Trained geometry-conditioned models over **frozen medical image features** for angle estimation; compared cross-attention and measurement-aligned sampling using five-fold out-of-fold landmark predictions, three seeds, and matched-versus-shuffled controls.

NextTier – IT Solutions Consultancy

Jun 2025 - May 2026

Machine Learning Engineer

Sacramento, CA (Remote)

- Built SQL-Agent, a **Qwen3-14B natural-language SQL agent** for registered SQLite data, with browser and FastAPI/MCP interfaces for revenue totals, order lists, grouped summaries, and result review.
- Engineered schema-aware **generate–execute–repair** loops with read-only authorization, query/output limits, and up to three attempts on a consistent data snapshot; routed general answers to human review.
- Built durable asynchronous execution with idempotent jobs, renewable worker leases, retries, and **stale-worker write protection**; validated recovery using worker-kill, restart, duplicate-request, and timeout tests.
- Achieved **91.0% correct runs (142/156) versus 65.4% (102/156)** for the single-pass Coder baseline in a configuration comparison on 26 development questions × 2 databases × 3 trials; used asynchronous delivery for 74.8 s p95 latency.

University of Missouri-Kansas City

Jan 2020 - May 2025

Graduate Research Assistant

Kansas City, MO

- Developed and trained transformer- and diffusion-based models for **thermal super-resolution, hyperspectral pansharpening, and NIR-to-RGB translation**; evaluated image quality and spectral fidelity.
- Co-authored a fully connected LSTM-CRF for clinical named-entity recognition, achieving **84.15% F1 on i2b2/VA**.

EDUCATION

University of Missouri-Kansas City

Ph.D., Electrical & Computer Engineering, GPA: 4.0/4.0

May 2025

M.S., Computer Science, GPA: 3.8/4.0

Dec 2019

PUBLICATIONS

8 peer-reviewed publications; 6 first-author imaging papers. Full list: Google Scholar.